



REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF AGRICULTURE
BUREAU OF AGRICULTURAL RESEARCH
 RDMIC Bldg., Elliptical Road corner Visayas Avenue,
 Diliman, Quezon City 1104

**DETAILED PROPOSAL
 (APPLIED/POLICY RESEARCH)**

I. BASIC INFORMATION	
1. Project Title	IMPROVING NUTRIENT BIOAVAILABILITY AND FEED QUALITY OF LOCALLY AVAILABLE ALTERNATIVE FEEDSTUFFS THROUGH FERMENTATION
2. Implementing Agency	
a. Lead Agency	CAPIZ STATE UNIVERSITY
b. Complete Address	CAPIZ STATE UNIVERSITY, BAILAN, PONTEVEDRA, CAPIZ
c. Agency Head	DR. EDITHA C. ALFON
d. Name and signature of the Project Leader	EMELY J. ESCALA, PhD
e. Name/s of the Study Leader/Project Staff	MANUEL D. GACUTAN JR., PhD STEPHANIE L. PIMENTEL, PhD JOSEPH F. DE LA CRUZ, PhD
f. Email Address	ejescala@capsu.edu.ph
g. Landline/Fax & Mobile Number	
3. Collaborating Agency(-ies)	Visayas State University University of the Philippines-Los Banos
4. Project Duration	2 years
5. Project Site/s	
a. Province	Capiz, Leyte, Laguna
b. City(-ies)/ Municipality (-ies)	
c. Barangay(s)	
d. Geocode(s)	
6. Project Funding	
a. Total Budget Requested	Php 5,744,853.60
b. Agency Counterpart	Php 1,014,080.00
7. Target Beneficiaries	
a. Next Users	Name/s: Number:
b. End Users	Name/s: Number:
8. RDE Agenda to be Addressed	Poultry and Livestock R4D: Animal Nutrition and Health



9. Expected Technology <i>(Please enumerate and refer to Annex B.2.)</i>	☒ Product:	1. Compounded Feed 2. Fermented Feed 3. Probiotics 4. Extracellular Products/secondary metabolites
	☒ Protocol/Process:	1. Fermented Feed Protocol 2. Feed Formulations 3. Characterization and isolation of probiotics 4. Extraction procedure of secondary metabolites or extracellular products (EXPs)
	☒ Information:	1. Biochemical, secondary metabolites, candidate probiotics 2. Policy on proper handling and processing of alternative feedstuffs
10. Technology Description		
This study will employ a transformative approach to sustainable animal nutrition by using locally available alternative nutrient resources reducing reliance on external inputs and other high-cost conventional feed ingredients. The development process will follow a scientific feed formulation and fermentation process to improve nutrient composition, palatability, digestibility, nutrient absorption and production of probiotics, and other significant metabolites or extracellular products (EXPs).		
11. Farm-level/Industry Application	1. Probiotics products as feed additive 2. Feed ingredients in dietary rations 3. Alternative to in-feed antibiotics and growth promoters	
12. Type of Research	☒ Applied ☒ Policy	

II. TECHNICAL DESCRIPTION
1. Rationale
Livestock, poultry, and aquatic products are the major contributors to global food security through the production of milk, eggs and meat. They are among the best sources of high-quality protein and micronutrients, both of which are essential for the normal growth and health of people. Livestock, poultry, and aquatic produced are also used to generate cash to purchase staple foods and other needs of the family. However, even with these benefits, it cannot be denied that they can contribute to food shortages by competing with humans for staple foods, particularly grains fed to pigs and poultry. There is a challenge in how to manage these complex trade-offs to realize livestock's and aquaculture's positive impacts while minimizing their negative effects (Smith et al., 2012). With the tremendous increase in population, livestock, poultry and aquaculture farmers are under unsustainable amounts of pressure to produce even more to meet the growing demand for animal and aquatic-based food products. However, several challenges must be overcome to meet this demand. One of the most important concerns is the global



scarcity of conventional feed resources as a result of many factors, such as increasing demand for grains for human consumption, industrialization, adverse weather conditions, rising crop-related diseases, and import policies. Consequently, the cost of conventional feed ingredients has been steadily rising over time. With the growing demand for animal and aquatic products as a result of globalization, and insufficiency and costly conventional feed materials, one possible and most practical alternative might be to include unconventional or alternative feedstuffs such as agricultural crop by-products (Sontakke et al., 2014; Amata, 2014), fisheries by-products, aquatic forage, and insect-based feed materials.

The Philippines is located in a tropical region. It is endowed with an abundance of feed resources encompassing terrestrial and aquatic origin. This climatic advantage fosters a diverse and flourishing agricultural landscape, making the country a habitat for a wide choices of alternative feedstuffs.

Nutrients derived from insects, foliage, crop residues, fish by-products are abundant but its feeding value are undermined. Instead of disposing these materials, repurposing them as animal and aquatic feed can be a cost-effective way to maximize its value. It will also reduce the accumulation of waste and promotes environmental sustainability by preventing them to be disposed into landfills or bodies of water thereby aligning them with the principles of circular economy and promoting renewable resource.

Since it is locally available, high cost of transportation and tariff are reduced. Further, by-products can provide a diverse range of nutrients which can contribute to meeting nutrient requirement of animals. However, these by-products are noted for their poor digestibility because of their innate anti-nutritional factors (ANFs) such as cellulose, hemicellulose and lignin (Swain et al., 2014), chitin, and other toxic metabolites. In light of recent reports, fermentation can reduce the fiber content of feed ingredients while also decreasing its ANFs (Sugiharto 2019) and toxins (Dai et al 2022).

Therefore, it is timely to investigate fermentation as a tool to improve the nutrient availability of these tropical feed resources.

This is an offshoot project from the previous study of the proponent funded by DA-BAR which was part of the Organic Agriculture Production Program of Capiz State University completed in 2018, wherein compounded and fermented feedstuffs were fed to the laying and growing Rhode Island Red chickens. Meanwhile, ten (10) alternative feedstuffs namely: water spinach, cassava tuber, rice bran, malunggay, soybean (whole), corn bran, oyster shells, telescope snails, crab carapace, mud clam, ark clam, shrimp meal, mungbean, sugarcane bagasse, coconut residue, and banana (pakol) have already been profiled in terms of its nutritive values. The feedstuffs to be used in this study will be added to the increasing inventory of potential locally available alternative feedstuffs in the country.

This will benefit farmers and fisherfolks in finding alternative feed resources based on their abundance in their respective localities and their prevailing farming systems. Further, different feedstuffs requiring different fermentation procedure and literatures are limited on the fermentation protocol for conventional feedstuffs, hence this study.

Lastly, this study is aligned with the United Nations SDGs 1 (no poverty), 2 (zero hunger), and among others. This is also in support with the DA-BAR and the regions research priority thrusts.



2. General Objective	Generally, this project aims to improve the nutrient bioavailability and feed quality of locally available alternative feedstuffs through fermentation.
3. Specific Objectives	1. Optimize the fermentation protocol for each feedstuffs; 2. Profile biochemical composition including secondary metabolites, and microbial consortia as probiotic candidate; 3. Conduct nutrient digestibility estimation and formulation of feed rations in livestock, poultry, and fisheries utilizing fermented alternative feedstuffs; 4. Evaluate growth performance and diet economics in livestock, poultry and fisheries utilizing fermented feedstuffs; 5. Investigate the effect of secondary metabolites or EXPs as feed additives; 6. Explore the probiotic effect of isolated microbial candidates in livestock, poultry, and fisheries; 7. Capacitate farmer-fisherfolks cooperators in the production and utilization of the fermented alternative feed resources; and 8. Craft a policy related to utilization of agri-and-fish by-products, and other alternative feedstuffs for sustained animal feeding and nutrition.

III. REVIEW OF RELATED LITERATURE
Insects and fish-c/o Manuel Agri by-products-c/o Probiotics and secondary metabolites Fermentation

IV. METHODOLOGY
Objective 1
Optimize the fermentation protocol for each feedstuffs. Identification, Collection and Preparation Among the commonly available crop and fisheries by-products with potential as feedstuffs are: a. Agricultural crop by-products



- i. Cassava (*Manihot esculenta*) foliage (most abundant variety in the region will be used)
- ii. Gabi (*Colocasia esculenta*)
- iii. Banana pseudostem
- iv. Rice straw
- v. Sweet potato leaves
- vi. Pineapple waste
- b. Aquatic by-products
 - i. shrimp waste
 - ii. fish waste
- c. Aquatic forage
 - i. azolla (*Azolla pinnata*)
 - ii. water hyacinth (*Eichhornia crassipes*)
 - iii. sargassum (*Sargassum muticum*)
- d. Insect feed material
 - i. Black soldier fly larvae
 - ii. Superworm larvae
 - iii. Meal worm larvae

These will be collected from the nearby farms and fish processing areas. Collected feedstuffs will be rinsed thoroughly to get rid of contaminants, air dried to 60-65% moisture, chopped to 2-3 cm particle size (Hsu et al 2013). Microbial inoculants such as cellulolytic yeast (*Entrophospora* sp.) or *Saccharomyces* sp, fungi (*Aspergillus oryzae*/ *A. niger*/ *Trichoderma reesei*), and bacteria (LAB strains) will be used as starter cultures (obtained at UPLB Biotech lab). A mixed liquid culture of isolates containing approximately 5 logcfu/ml, will be prepared in a 1:1 ratio. This will be inoculated at ten percent (10%) v/w liquid starter culture (1×10^7 CFU/g) into the substrate then pack in multi-layer polyethylene bags to permit anaerobic fermentation with gas-pressure opening valve and incubated at 30-40°C (Liu et al., 2010a, Liu et al., 2010b, Wang et al., 2014a, Wang et al., 2014b) until constant pH and temperature will be attained.

To facilitate temperature and pH monitoring, the cover of the container will be installed with thermometer and canula for the collection of the pH. The pH 6 is a good initial level to start the fermentation process, while the pH of the fermented products should fall under pH 3-5 (Vatansever et 2017, Engberg et al 2009).

Fish waste ensiling will be done using the procedure of Toppe et al (2018), with some modifications.

Objective 2

Profile biochemical composition including secondary metabolites, and microbial consortia as probiotic candidate



To quantify the biochemical composition of the fermented alternative feeds, samples will be pooled, subjected to appropriate chemical assays including proximate analysis, energy determination, mineral analysis, and ANFs assays in collaborating agencies, and commercial laboratories.

For microbial isolates, a 100g representative sample of the pre-and-post fermented substrates will be pounded. One (1g) gram will be added to 9 ml of PBS for serial dilution. Genomic DNA will be extracted using HiPurA™ Bacterial Genomic DNA Purification Kit. Extracted DNA will be sent to Apical Scientific for shotgun metagenomic sequencing using Illumina platform. Metagenomic sequence results will be analyzed using the pipeline: quality check, trimming, mapping, search against probiotic database and construction of pie graph using Krona software.

Objective 3

Conduct nutrient digestibility estimation and formulation of feed rations in livestock, poultry, and fisheries utilizing fermented alternative feedstuffs;

Fermented alternative feedstuffs will be served as test ingredient in digestibility trials. Adjustment period will be appropriately observed before the introduction of the test diets in livestock (goat), poultry (Native chicken), and fisheries (catfish). Fasting requirement will be observed prior to forced feeding of known amount of the test diet. Marker-to-marker approach using TiO_2 will be used to determine start of feeding and end of excreta collection. Calculations of metabolizable energy will be calculated using total tract digestibility using the formula:

$$ME = \frac{\text{Feed Energy} - \text{Excreta Energy}}{\text{Feed Energy}} \times 100$$

Formulation of feed ration will be done based on the nutrient composition **based** on specific animal requirement with the suggested inclusion rate of the fermented alternative feedstuffs

Objective 4

Evaluate growth performance and diet economics in livestock, poultry and fisheries utilizing fermented feedstuffs;

GOAT and Chicken

Formulated diets will be provided to post weaned goats (at least 3 mos old), and/or chicken (hardened), and growth performance will be evaluated based on body weight (BW), voluntary feed intake (VFI), dry matter intake (DMI), weight gain (WG), and feed conversion ratio (FCR). Return above feed cost will also be calculated to determine diet economics of utilizing fermented alternative feed resources.

The trial will compare compounded Fermented Feed and Non-Fermented Feed. The experiment will be laid out in a Randomized Complete Block Design with initial weight and sex of the the animal as blocking factor. It will be blocked six (6) times with subsamples (10 for birds and 2 for goat). Data will be checked for homogeneity of variances using Bartlett's test and normality using Wilk Shapiro's test and remedial measures will be done if necessary. Analysis of variance in RCBD will be run using Proc GLM of SAS version 4. Pairwise test using Tukey's HSD will be used to compare paired treatment means and will be declared significant at $p < 0.05$.

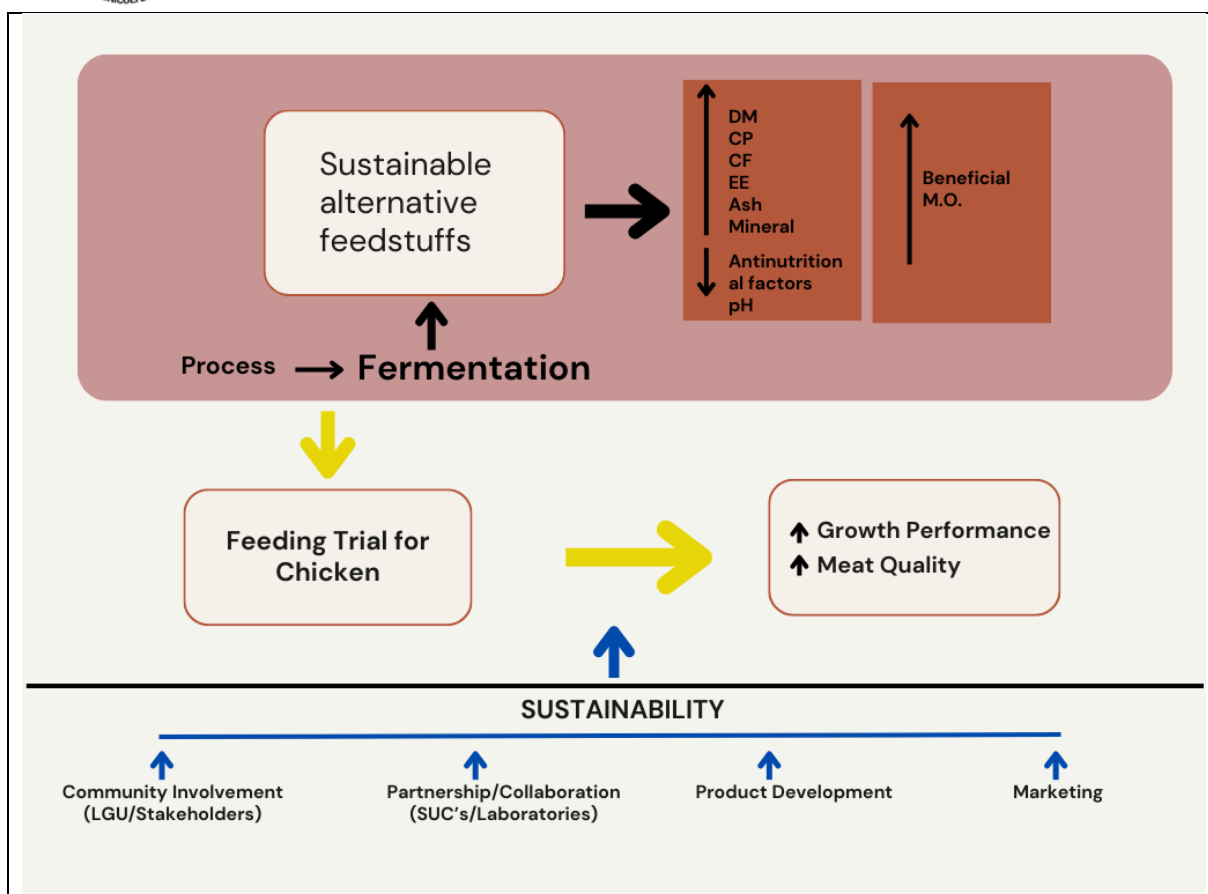
This will be conducted at the Livestock and Poultry Farm of CapSU Pontevedra.



Catfish
Objective 5
Investigate the effect of secondary metabolites or EXPs as feed additives
Objective 6
Explore the probiotic effect of isolated microbial candidates in livestock, poultry, and fisheries; Probiotic candidate will be isolated, cultured as a feed additive to chicken. Intestinal morphology and immune health will be determined as an effect of the probiotic feed additive.
Objective 7
Capacitate farmer-fisherfolks cooperators in the production and utilization of the fermented alternative feed resources; and
Objective 8
Craft a policy related to utilization of agri-and-fish by-products, and other alternative feedstuffs for sustained animal feeding and nutrition. Create a collaboration between CapSU, LGUs, Fisher folks, and Farmers' associations to provide an avenue for the information dissemination and encourage active participation of the different stakeholders. Subsequently, conduct series of orientation on the importance of crop and fisheries by-products, foliage and aquatic plants on the improvement of a sustainable and resilient livestock and poultry production, as well as skills training on proper handling of the feedstuffs for fermentation.
Conceptual Framework



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Expected Outputs (EO)	
Output per Objective	6Ps (Product, Publications, People and Services, Places and Partnerships, Patent/Intellectual Property, Policy)
EO 1: Fermentation protocol for each alternative feed resources	Patent/Intellectual Property- Fermentation protocol
EO 2: Information on the biochemical composition, secondary metabolites, and probiotic candidates	Publication- Information on the biochemical composition, secondary metabolites, and probiotic candidates
EO 3: Digestibility estimates of nutrients and develop feed formulation	Publication- Information on nutrient digestibility Product-single and compounded ration, fermented feeds
EO 4: Data on the animal performance and diet economics	Publication- Information on the effect of feeds on the animals



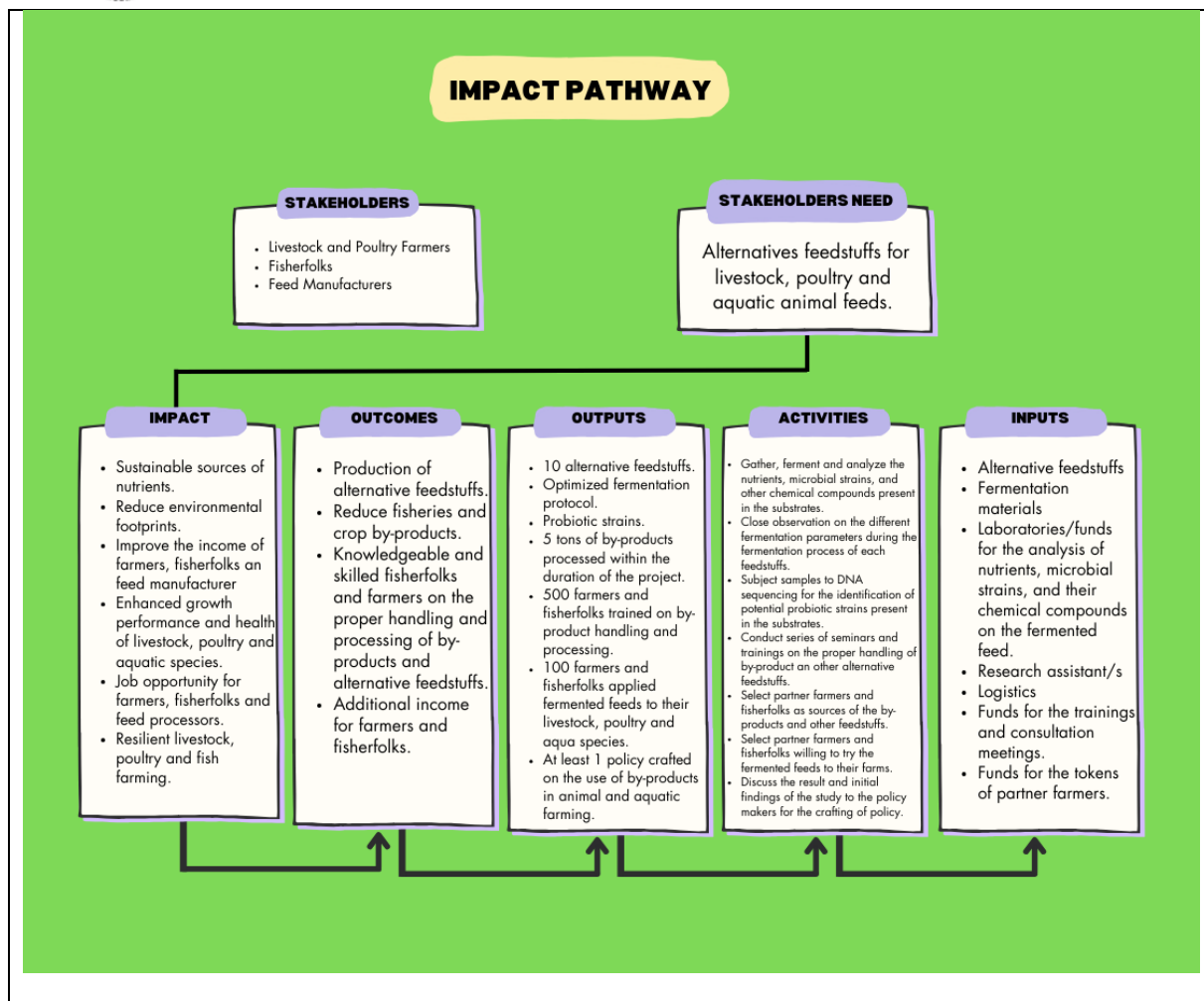
	- Information on the cost of feed
EO 5: Information on the secondary metabolites and/or EXPs present in the fermented alternative feed resources	Publication-Information on the concentration of the secondary metabolites or EXPs as feed additives Protocol- Extraction procedure of the secondary metabolites and EXPs Product-secondary metabolites/EXPs
EO 6: Information on the mode of action of the probiotic as feed additive in chicken	Publication-Information of the probiotic effect in chicken Protocol-Extraction and dose administration of probiotic as a feed additive Product-Probiotic
EO 7: Trained farmer-fisherfolk cooperators in the production and utilization of fermented alternative feedstuffs	People and Services-500 farmers-fisherfolk cooperators
EO 8: Policy on the utilization of agri-and-fish by-products, and other alternative feedstuffs	Policy- Policy on the utilization of agri-and-fish by-products, and other alternative feedstuffs People and Services-at least 20 farmers and fisherfolks adopted the technology
Outcome	• Production of alternative sources of feeds for the livestock and poultry raisers in the province of Capiz. • Reduce fisheries and crop by-products. • Knowledgeable and skilled fisherfolks and farmers on the proper handling and processing of by-products. • Additional income for farmers and fisherfolks for the sales of their by-products.
Impact	• Sustainable sources of alternative feeds for livestock and poultry farmers. • Reduction of environmental footprints associated with the disposal of crop and fisheries by-products.



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	• Additional income for the farmers, fisherfolks and fermented feed processors. • Enhanced overall productivity of livestock as an effect of improved feed formulation using fermented alternative feedstuffs. • Job creation as an offshoot to the development of industries focusing on processing and utilizing by-products. • Resilient livestock and poultry production system for farmers as they diversify their feed resources, reduce vulnerability to fluctuations in the prices of conventional feeds as they have options during periods of scarcity and market disruptions.
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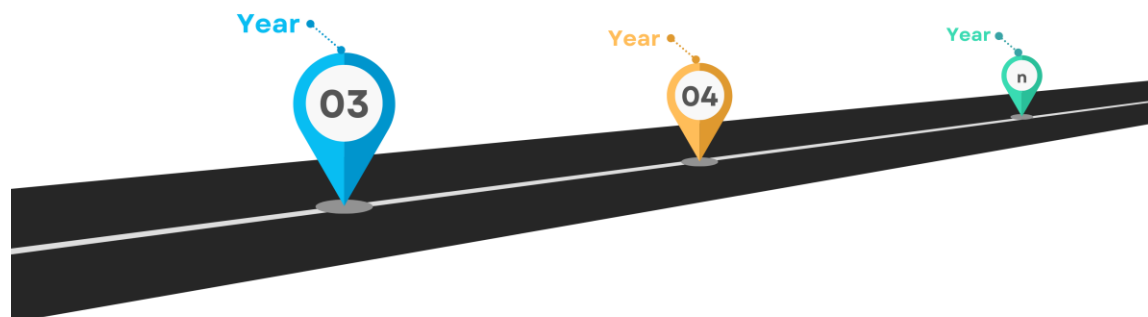
V. IMPACT PATHWAY



VI. EXIT/SUSTAINABILITY PLAN



SUSTAINABILITY PLAN



FINANCIAL MANAGEMENT

Defining cost per kilogram of by-products and other alternative feedstuff based on their production, collection and other handling expenditure.

Conducting quality and safety checks of the feedstuffs to meet standards and increase market quality.

Forming strategic partnership with local farmers and fisherfolks, and feed businesses, or agricultural cooperatives to secure a consistent and stable supply of raw materials.

Collaborating with other research institutions, NGO, or other funding agencies, government institutions to leverage resources expertise.

Implementing efficient production processes to minimize production cost.

Allocating funds for product improvement and promotion.

Exploring other grants, subsidies, or incentives to fund for the start-ups.

Commercialization of fermented alternative feedstuffs.

SOCIAL ASPECTS

Fostering community involvement in the project to gain support for the continuous production of raw materials and adoption of technology and products.

Fostering community involvement in the project to gain support for the continuous production of raw materials and adoption of technology and products.

Providing training and skill development activities to enhance the capacity of the local workforce towards the production and sustainable alternatives feedstuffs.

Developing educational programs on sustainable farming practices, resource management, and environmental conservation related to the production of alternative feed resources and its application to livestock, poultry and fish farming.

Considering small-scale farmers or cooperatives in the crafting of the business model.

Improving health and nutrition of the community through the production of healthy and well-fleshed meat, egg and fish products.

Strong collaboration with NGO's, community leaders, and government agencies to aligning the project activities with community needs and priorities.

ENVIRONMENTAL

Drafting of a policy aimed at guiding the repurposing of crop and fisheries by-products.

Persistent reduction of crop and fisheries by-products.

Reducing carbon footprint from the disposal of crop and fisheries by-products.



VII. POTENTIAL INTELLECTUAL PROPERTY	
1. Type	☒ Patent ☐ Trademark ☐ Utility Model ☒ Copyright
2. Status	☐ None ☐ For application ☐ With pending application ☐ Registered
3. Description	

VIII. LOGICAL FRAMEWORK
<i>(see attached)</i>

IX. WORK PLAN SCHEDULE
<i>(see attached)</i>

X. BUDGET SUMMARY AND WORKSHEET DETAILS
<i>(see attached)</i>

XI. LITERATURE CITED
Amata, I. (2015). The use of non-conventional feed resources (NFCR) for livestock feeding in the tropics: A Review. <i>Journal of Natural Product and Plant Resources</i> 5,7-15. Association of Analytical Chemists. (2022). <i>Official Methods of Analysis</i>. Arlington Dai, Z., Cui, L., Li, J., Wang, B., Guo, L., Wu, Z., Zhu, W., Wu, G. (2020). Chapter 24 - Fermentation techniques in feed production. <i>Animal Agriculture</i>, Academic Pages 407-429, ISBN 9780128170526, https://doi.org/10.1016/B978-0-12-817052-6.00024-0 Engberg, R.M., Hammershoj, M., Johansen, N.F., Abousekken, M.S., Steenfeldt, S., and Jensen, B.B. (2009). Fermented feed for laying hens: effects on egg production, egg quality, plumage condition and composition and activity of the intestinal microflora. <i>British Poultry Science</i>, 50:2, 228-239, DOI: 10.1090/00071660902736722. Escala, E., Barroga, A., San Andres, J., Rafael, R., Undan, J., Latonio, A. Mingala, C. (2022). Influence of fermentation method on the nutrient composition of locally available feedstuffs in the Philippines. A Dissertation. Hsu et al., 2013 Liu et al., 2010a Liu et al., 2010b



Smith et al., 2012

Sontakke, U., Kale, V., Bose, B., Kumar, M., (2014). Non-conventional Feeds and Agro Industrial By-products: Their scope and Future Demand for Livestock Production. Dairy Cattle Nutrition Division. National Dairy Research Institute.

Sugiharto, S. and Samir, R. (2019). Recent advances in fermented feeds towards improved broiler chicken performance, gastrointestinal tract microecology and immune responses: A review. Animal Nutrition. Vol. 5, Issue 1, pages 1-10, ISSN 2405-6545, <https://doi.org/10.1016/j.aninu.201811.001>.

Swain et al., 2014

Toppe et al., 2009

Vatansever, S., VEGI, A., ROBINSON, J., Hall, C. (2017). The Effect of Fermentation on the Physicochemical Characteristics of Dry-Salted Vegetables. Journal of Food Research. Vol 6, no. 5. Canadian Center of Science and Education. ISSN 1927-0887.

Wang et al., 2014a

Wang et al., 2014b



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ANNEX B.1. GUIDELINES FOR THE PREPARATION OF PROPOSAL FOR APPLIED AND POLICY RESEARCH

Detailed proposals must be prepared using the format given and in accordance with the following guidelines. Use Cambria 11, single space.

I. BASIC INFORMATION

<i>Project Title</i>	Brief but clear official and distinctive name of the project that describes the main theme of the proposed study
<i>Implementing Agency</i>	Primary institution that will lead the implementation of the project
<i>Lead Agency</i>	Name of the institution where the project leader is affiliated
<i>Complete Address</i>	Indicate the block and lot number, street, subdivision, barangay, city/municipality, province, postal code
<i>Agency Head</i>	Indicate the Last Name, First Name, Middle Name of the head of the agency
<i>Name and Signature of the Project Leader</i>	Principal researcher or implementer of the project. Indicate the Surname, First Name, Middle Name and position/designation
<i>Name/s of the Study Leader/ Project Staff</i>	Indicate the Last Name, First Name, Middle Name and position/designation of the project staff
<i>Email Address</i>	Email address of the project leader
<i>Landline/Fax & Mobile Number</i>	(Area code) NNNN-NNNN loc NNN 09XX-XXX-XXXX
<i>Collaborating Agency(-ies)</i>	Agency (-ies) or FCAs that will be cooperating with the lead agency implementing the project
<i>Project Duration</i>	Period that indicates proposed start and completion of the project (in number of months or years)
<i>Project Site/s</i>	Place or area covered by the project or where the project will be conducted
<i>Project Funding</i>	Financial requirement of the project in Philippine Peso
<i>Total Budget Requested</i>	Total budget requested from DA-BAR



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<i>Agency Counterpart</i>	The contribution/ share of the implementing/collaborating agency in the conduct of the project
<i>Target Beneficiaries</i>	Groups or individuals who will benefit from the conduct of the project. Include the specific target users and estimated number of beneficiaries
<i>Next Users</i>	Enablers for the access and promotion of technologies to the end-users (ie. NGAs, LGUs). Specify the name/s and number of beneficiaries (individual or groups) of the project
<i>End Users</i>	Technology adoptors such as farmers and fisherfolk. Specify the name/s and number of beneficiaries (individual or groups) of the project
<i>RDE Agenda Addressed</i>	Researchable area to be addressed by the project based on DA Thrusts and Priorities, RIRDEAP or DA-BAR's NAFRDEAP
<i>Expected Technology</i>	Knowledge and know-how, skills, products, processes, and/or practices expected to be generated from the project. Enumerate the specific technology/ies under product, protocol/process, or information. Refer to <i>Annex B.2</i> .
<i>Technology Description</i>	Briefly describe the technology that will be developed from the conduct of the project
<i>Farm-level/Industry Application</i>	Indicate the immediate application of the project to the farmer- and fisherfolk-level and the industry
<i>Type of Research</i>	Indicate whether applied or policy type of research

II. TECHNICAL DESCRIPTION

<i>Rationale</i>	Definition of research problem, scope and coverage, significance of the research project and the justification for its conduct. Include baseline data, statistical data, scientific findings, and among others. Briefly discuss the alignment of the project to the DA thrusts and priorities, its relevance, and merits. Provide a hypothesis or the answer/s to problems and the interventions/technologies to be made.
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<i>General Objective</i>	General purpose of the project to address the problems/researchable areas and must be consistent with the title
<i>Specific Objectives</i>	Enumerate specific, measurable, attainable, realistic, and time-bound objectives
III. REVIEW OF RELATED LITERATURE	Concepts and approaches gathered from recent literature which indicate gaps in existing knowledge to help determine direction and focus of the proposed project and avoid possible duplication. Please cite references.
IV. METHODOLOGY	Indicates the project components (if applicable), and contains the details and procedure to be followed in conducting the research. This includes: 1. Factors (variables) on the experiment or study 2. Treatments and layout 3. Experimental design, replications, characteristics of experimental units (sites, number, area, etc.) 4. Statistical analysis 5. Specific management of the experiment (includes specific features about the management of the experiment that are not included in the treatments) 6. Cultural practices (ie. land preparation, pest control, weed control, fertilization)
<i>Conceptual Framework</i>	Illustrate the expected relationships among project components/activities towards the attainment of the overall objective.
<i>Expected Output</i>	Indicate the projected achievement/milestones/significant findings/specific product or services that will be generated upon completion of a particular activity. Provide output per objective following the 6Ps metrics.
<i>Outcome</i>	Short to medium term effects of the project to its intended beneficiaries after its successful completion.
<i>Impact</i>	Long term effects of the project to its intended beneficiaries after its successful completion in terms of social, environmental, and economic impact.



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V. IMPACT PATHWAY

Describe the pathway/plans/activities on how the potential impact will be achieved/delivered to intended beneficiaries. Consider the theory of change, effect of external factors, stakeholders to be engaged, as well as the monitoring and assessment activities.

**VI. EXIT/
SUSTAINABILITY PLAN**

Indicate the strategies to be made, by the end of the project, to overcome potential financial, social, and environmental risks in order to ensure the continuity of the intervention/s.

**VII. POTENTIAL
INTELLECTUAL
PROPERTY APPLICATION**

Specifies the expected results from the project that are patentable such as new inventive steps that have commercial application. Indicate and describe whether patent, utility model, trademark, or copyright.

**VIII. LOGICAL
FRAMEWORK**

Presents the main elements of the project, expressed in terms of objectives for each element and inputs required to achieve them

**IX. WORK PLAN
SCHEDULE**

Itemizes the detailed activities to be undertaken for the whole duration of the project

**X. BUDGET SUMMARY
AND WORKSHEET
DETAILS**

Describes the personnel/staff and material resources needed by the project; provides information on human and non-human resource requirements of the project expressed in budgetary terms. It consists of four forms: Budget Summary, Worksheet Details for Personnel Services (PS), Maintenance and Other Operating Expenses (MOOE) and Equipment Outlay (EO).



ANNEX B.2. TYPES OF TECHNOLOGY/RESEARCH OUTPUTS

1. Product (tangible end outputs from the R4D activities)

a. Processed food products

Example: Ready-to-Drink (RTD) juice and preserves

Product:

1) RTD juice from passion fruit; 2) RTD juice from lipote; 3) RTD juice from bignay; 4) Preserves from passion fruit; 5) Preserves from lipote; 6) Preserves from bignay

Processing technology:

7) Processing technology for juice
8) Processing technology for preserves

b. Non-food products

1. Consumables

i. Feeds
ii. Fertilizers, inoculants, soil amendments
iii. Inoculants
iv. Pesticides, herbicides, insecticides
v. Cultivar or variety
vi. Textiles, packaging materials

2. Non-consumables

i. Machine/equipment
ii. Test kits
iii. Web/mobile applications, information systems, database platforms

2. Process (non-tangible end outputs from the R4D activities)

- a. Production technology/protocol (eg. tissue culture, fertilizer formulation and production, cultural management practices [*input- and production-related*])
- b. Postharvest technology/protocol (eg. hot water treatment for mango, storage protocols to prolong shelf life)
- c. Processing technology/protocol

3. Knowledge/Information

- a. Policy recommendation (value/supply chain, assessment reports [impact, ex ante, ex post])
- b. Marketing and business strategies
- c. Decision support tool, specifically information generated from various applications/platforms (eg. SSNM, PRiSM)